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R3.2 Electron transfer reactions

# Oxidation and reduction

## Learning outcomes

By the end of this section you should be able to:

- (R3.2.1) Deduce the oxidation state of an atom in a compound or ion.
- (R3.2.1) Identify the oxidised and reduced species and the oxidising and reducing agents in a chemical reaction.
- (R3.2.2) Deduce redox half-equations and equations in acidic or neutral solutions.

**Figure 1.** A rusty bicycle.

Credit: fhm, Getty Images

You might have noticed that certain metals rust with time. For example, the bicycle in **Figure 1** is covered in that characteristic reddish-brown colour of rust which indicates the presence of iron(III) oxide, produced when iron reacts with oxygen in the air:

$$4Fe(s) + 3O_2(g) \rightarrow 2Fe_2O_3(s)$$

When any iron material is left in contact with air, you will see it rust. You might have heard this described as the oxidation of iron.

# Defining oxidation and reduction

The rusting of iron can be represented by the equation:

$$4Fe(s) + 3O_2(g) \rightarrow 2Fe_2O_3(s)$$

## Worked example 1

Describe what happens to iron in the equation:

$$4Fe(s) + 3O_2(g) \rightarrow 2Fe_2O_3(s)$$

Iron, Fe, reacts with oxygen to form a new compound, iron(III) oxide, $Fe_2O_3$. It can be said that the iron gains oxygen, or becomes oxidised.

In short, iron undergoes oxidation.

Use **Interactive 1** to assign oxidation states (see section S3.1.6 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/oxidation-states-id-44696/)) to all the species involved in the rusting of iron.

$$4Fe(s) + 3O_2(g) \rightarrow 2Fe_2O_3(s)$$

| | | |
|---|---|---|
| +7 | +8 | +9 |
| +4 | +5 | +6 |
| +1 | +2 | +3 |
| | 0 | |
| −1 | −2 | −3 |
| −4 | −5 | −6 |
| −7 | −8 | −9 |

Check

**Interactive 1.** Assigning Oxidation States.

## Worked example 2

For the rusting of iron:

$$4Fe(s) + 3O_2(g) \rightarrow 2Fe_2O_3(s)$$

describe what is happening to iron and to oxygen in terms of their oxidation states.

The oxidation state of iron is *increasing* from zero to +3. The oxidation state of oxygen is *decreasing* from zero to –2.

It can be said that:

- Iron is increasing in oxidation state or becoming oxidised. Iron is undergoing oxidation.

- Oxygen is decreasing in oxidation state or becoming reduced. Oxygen is undergoing reduction.

You can also consider what happens to iron and oxygen in terms of the number of electrons each possess, based on the changes to their oxidation states.

## Worked example 3

For the rusting of iron:

$$4Fe(s) + 3O_2(g) \rightarrow 2Fe_2O_3(s)$$

describe what happens to iron and to oxygen in terms of the number of electrons that each species possesses.

Each iron atom loses three electrons and becomes a cation, $Fe^{3+}$.

Each oxygen atom gains two electrons and becomes an anion, $O^{2-}$.

It can be said that:

- Iron loses electrons or becomes oxidised. Iron undergoes oxidation.
- Oxygen gains electrons or becomes reduced. Oxygen undergoes reduction.

Analysing all the information above, you can see that oxygen was added to iron – this is called oxidation. Whenever one of the species in the chemical reaction is oxidised, another species is reduced. It is said that it undergoes reduction, which in this case could be defined as the removal of oxygen. The definition of oxidation as the addition of oxygen is historical, not all oxidation-reduction reactions involve the addition of oxygen.

### Making connections

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The surface oxidation of metals is often known as corrosion. What are some of the consequences of this process? (see subtopic S2.3 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43111/)).

Rusting is an example of corrosion. Corrosion creates structural damage and reduces the functionality of the material. It can lead to increased maintenance costs, safety risks, economic impact and reputational damage. The use of preventive measures, such as proper material selection, protective coatings and regular maintenance, can help mitigate the consequences of corrosion.

Photosynthesis is a reduction reaction in which plants use energy from the Sun to convert carbon dioxide and water into glucose (sugar) and oxygen, according to the equation:

$$6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$$

Describe what happens to carbon dioxide during photosynthesis.

Carbon dioxide, $CO_2$, reacts with water to form a new compound, glucose, $C_6H_{12}O_6$. It can be said that the carbon atom gains hydrogen, or becomes reduced. Carbon undergoes reduction.

Describe what happens to carbon and to oxygen in terms of their oxidation states during photosynthesis.

$$\overset{+4\;-2}{6CO_2} + \overset{+1\;-2}{6H_2O} \rightarrow \overset{0\;+1\;-2}{C_6H_{12}O_6} + \overset{0}{6O_2}$$

The oxidation state of carbon decreases from +4 to zero.

The oxidation state of oxygen increases from –2 to zero.

It can be said that:

- Carbon decreases in oxidation state or becomes reduced. Carbon undergoes reduction.

- Oxygen increases in oxidation state or becomes oxidised. Oxygen undergoes oxidation.

Based on the changes to their oxidation states, describe what happens to carbon and oxygen in terms of the number of electrons they possess.

Each carbon atom gains four electrons and each oxygen atom loses two electrons. In this case, no ions are created as carbon and oxygen share their electrons in covalent bonds.

It can be said that:

- Carbon gains electrons or becomes reduced. Carbon undergoes reduction.
- Oxygen loses electrons or becomes oxidised. Oxygen undergoes oxidation.

You can see hydrogen atoms were removed from the water (oxidation) and added to the carbon dioxide (reduction) – this is another way to define redox reactions.

Recall that ions are formed when an atom loses or gains electrons (see subtopic S2.1 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43108/)).

The reaction of zinc metal with copper(II) ions in a $CuSO_4$ solution has the equation:

$$Zn(s) + CuSO_4(aq) \rightarrow ZnSO_4(aq) + Cu(s)$$

Notice that zinc metal, Zn(s), loses electrons (oxidation), becoming zinc ions, $Zn^{2+}$(aq), while copper ions, $Cu^{2+}$, gain electrons (reduction) becoming copper metal, Cu(s). The sulfate ions, $SO_4{}^{2-}$, remain unchanged – they are called spectator ions.

You can rewrite the equation, omitting the spectator ions as they are not involved in the reaction.

Rewrite the reaction of zinc and copper(II) sulfate, omitting the spectator ions.

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$$Zn(s) + Cu^{2+}(aq) \rightarrow Zn^{2+}(aq) + Cu(s)$$

The copper ions in the copper(II) sulfate compound have a charge of 2+. This means that the zinc ion in the zinc sulfate also has a charge of 2+.

Assign oxidation states to all of ions in the previous equation and determine which species were oxidised and which were reduced.

$$\overset{0}{Zn}(s) + \overset{+2}{Cu^{2+}}(aq) \rightarrow \overset{+2}{Zn^{2+}}(aq) + \overset{0}{Cu}(s)$$

The oxidation state of zinc increases from zero to +2.

The oxidation state of copper decreases from +2 to zero.

It can be said that:

- Zinc increases in oxidation state or becomes oxidised. Zinc undergoes oxidation.
- Copper decreases in oxidation state or becomes reduced. Copper undergoes reduction.

Look at the equation that represents the reaction that transforms methanol, $CH_3OH$, to methanal, HCHO:

$$6CH_3OH + Cr_2O_7^{2-} + 8H^+ \rightarrow 6HCHO + 2Cr^{3+} + 7H_2O$$

One of the most common oxidising agents, potassium dichromate, $K_2Cr_2O_7$, is used in the reaction. You will learn more about oxidising agents later in this section. The reaction takes place in the presence of acid ($H_2SO_4$) and heat.

## Worked example 4

Deduce the oxidation states of all atoms in the equation above and determine which species are oxidised and reduced.

$$\overset{-2\ +1\ -2}{6CH_3OH} + \overset{+6\ -2}{Cr_2O_7^{2-}} + \overset{+1}{8H^+} \rightarrow \overset{+1\ 0\ -2}{6HCHO} + \overset{+3}{2Cr^{3+}} + \overset{+1\ -2}{7H_2O}$$

The oxidation state of carbon increases from –2 to 0.

The oxidation state of chromium decreases from +6 to +3.

It can be said that:

- Carbon increases in oxidation state or becomes oxidised. Carbon undergoes oxidation.
- Chromium decreases in oxidation state or becomes reduced. Chromium undergoes reduction.

## Study skills

Include the use of oxidation numbers in the naming of compounds, as in the examples above for iron(III) oxide and copper(II) sulfate.

Note that there is no space between the name of the metal and the parentheses that include the Roman numeral for the oxidation state.

Recall that the terms 'oxidation number' and 'oxidation state' are often used interchangeably, and either term is acceptable in assessment (see section S3.1.6 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/oxidation-states-id-44696/)).

There are four ways to define oxidation and reduction. Complete the table in **Interactive 2** by dragging the correct sentences to the spaces to create correct comparisons.

| | Oxidation | Reduction |
|---|---|---|
| **Oxidation state** | | |
| **Electron transfer** | | |
| **Oxygen** | | |
| **Hydrogen** | | |

Added

Decreases

Removed

Gain

Increases

Removed

Loss

Added

Check

**Interactive 2.** Summary Table for the Definitions of Oxidation and Reduction.

## Making connections

What are the advantages and limitations of using oxidation states to track redox changes? (see subtopic S3.1 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/content-to-come-id-43116/)).

Oxidation states are a useful tool for tracking redox changes in many chemical reactions. They provide a simple and systematic way to track changes in electron transfer during redox reactions. Oxidation states follow specific rules and guidelines and can help students understand and visualise the changes in electron distribution during redox reactions, aiding in conceptual understanding.

However, they have limitations and should be used with caution, especially in complex chemical systems, such as enzymes, organometallic compounds and transition metal complexes. Oxidation states are an approximation of the electron distribution in a compound or ion, and they may not always accurately reflect the true electron density and distribution. Careful consideration of the specific context and chemical environment is necessary for an accurate interpretation of oxidation states in redox reactions.

# Oxidising and reducing agents

In redox reactions, the reduction of one species occurs only because there is oxidation of the other species. Therefore, we can say that:

- the species that causes oxidation is the oxidising agent and becomes reduced.
- the species that causes reduction is the reducing agent and becomes oxidised.

**Figure 2** shows a representation of the roles of oxidising and reducing agents.

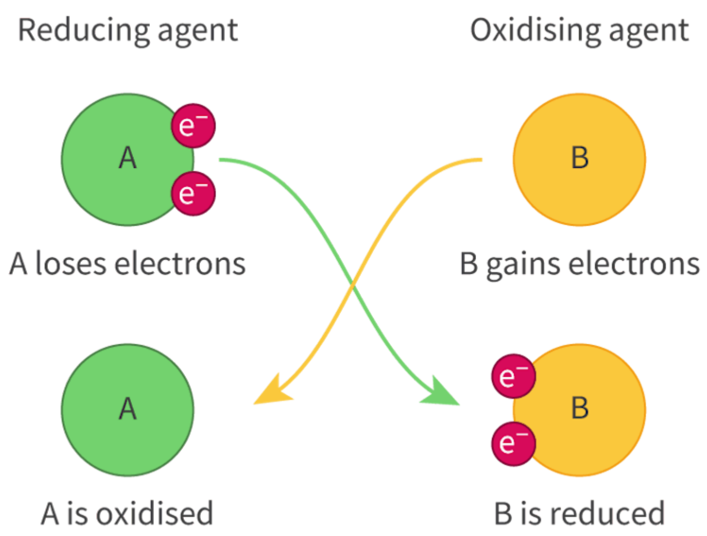


**Figure 2.** The different roles of oxidising and reducing agents in terms of electron transfer.

More information for figure 2

The diagram illustrates the roles of oxidizing and reducing agents with electron transfer between two species labeled as A and B. On the left, a green circle labeled 'A' represents the reducing agent, which loses electrons (e-) as indicated by two small red circles labeled with 'e-'. The process is depicted by a green arrow pointing from A to the electrons. Below this, it states 'A is oxidised'.

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On the right, a yellow circle labeled 'B' is shown as the oxidizing agent, which gains the electrons that A loses. The gained electrons are illustrated with red circles attached to B, and an orange arrow points towards it. Below this, the text reads 'B is reduced'.

The diagram visually represents electron transfer where A loses and B gains, thereby showing A is oxidized and B is reduced.

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Let’s apply this to the reactions below.

## Worked example 5

Deduce the oxidation states of the atoms in the reaction below, when copper(I) oxide is heated in air or oxygen:

$$2Cu_2O + O_2 \rightarrow 4CuO$$

$$\overset{+1}{2Cu_2}\overset{-2}{O} + \overset{0}{O_2} \rightarrow \overset{+2}{4Cu}\overset{-2}{O}$$

The copper ion attached to oxygen in the reactants has a charge of +1, given that there are two copper ions bonded to one oxygen ion with a charge of 2–.

The copper ion attached to oxygen in the products has a charge of +2, given that there is only one copper ion bonded to one oxygen ion with a charge of 2–.

## Worked example 6

Based on the oxidation states found in the previous question, identify the oxidising and reducing agents when copper(I) oxide is heated in air or oxygen:

$$2Cu_2O + O_2 \rightarrow 4CuO$$

The oxidising agent in this reaction is oxygen.

The oxidising agent leads to the oxidation of the other species and becomes reduced (decreased in oxidation state).

The reducing agent in this reaction is copper.

The reducing agent leads to the reduction of the other species and becomes oxidised (increased in oxidation state).

Identify the oxidising agent in the iron rusting reaction:

$$4Fe(s) + 3O_2(g) \rightarrow 2Fe_2O_3(s)$$

The oxidising agent in this reaction is the oxygen atom.

The oxidising agent leads to the oxidation of the other species and becomes reduced.

Identify the reducing agent in the iron rusting reaction:

$$4Fe(s) + 3O_2(g) \rightarrow 2Fe_2O_3(s)$$

The reducing agent in this reaction is the iron atom.

The reducing agent leads to the reduction of the other species and becomes oxidised.

In this reaction, iron increased in oxidation number from zero to +3, which means it was oxidised and, therefore, caused the reduction of oxygen (that decreases in oxidation number from zero to −2).

# Writing redox equations

As described previously, if you place zinc metal, Zn(s), in a solution of copper(II) sulfate, the zinc metal loses electrons, becoming zinc ions, $Zn^{2+}$(aq), while the copper ions, $Cu^{2+}$, gains electrons becoming copper metal, Cu(s).

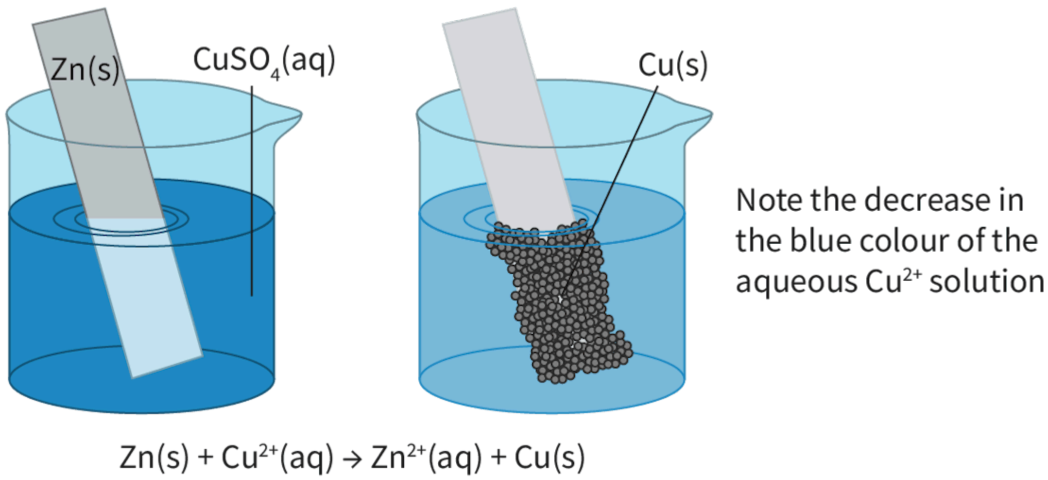


**Figure 3.** The reaction of zinc metal with copper(II) sulfate solution.

More information for figure 3

The image shows an illustration of a chemical reaction between zinc metal and copper(II) sulfate solution in two stages, represented by two beakers. The first beaker on the left contains a blue solution labeled "$CuSO_4$(aq)," with a zinc strip labeled "Zn(s)" partially submerged. The second beaker on the right shows the same setup but with the zinc strip now covered in small black particles labeled "Cu(s)," indicating copper metal formation. Below the beakers is the reaction equation: "Zn(s) + $Cu^{2+}$(aq) → $Zn^{2+}$(aq) + Cu(s)." Additional text notes a decrease in the blue color of the solution, which illustrates the reduction of copper ions to copper metal and the oxidation of zinc to zinc ions.

[Generated by AI]

How can we represent what happens in oxidation and reduction?

For any redox reaction, we can write separately what happens to each of the species in two half-reactions. Zinc metal, Zn(s), lost electrons and became oxidised, which can be represented by:

Oxidation:

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$$Zn(s) \rightarrow Zn^{2+}(aq) + 2e^-$$

Note that each zinc atom loses two electrons ($e^-$) that are represented on the products side. Copper ions, $Cu^{2+}$, receive electrons and are reduced, becoming copper metal, Cu(s).

Reduction:

$$Cu^{2+}(aq) + 2e^- \rightarrow Cu(s)$$

Note that each copper ion gains two electrons ($e^-$) that are represented on the reactants side.

Now it is your turn to do the same for the iron rusting reaction:

$$4Fe(s) + 3O_2(g) \rightarrow 2\ Fe_2O_3(s)$$

Start by identifying the two ions in iron(III) oxide and focus on describing what happens to each of them.

# Worked example 7

State the oxidation half-reaction for the rusting of iron.

Oxidation:

$$Fe(s) \rightarrow Fe^{3+}(s) + 3e^-$$

Each iron atom loses three electrons and forms an iron(III) ion.

# Worked example 8

State the reduction half-reaction for the rusting of iron.

Reduction:

$$O_2(g) + 4e^- \rightarrow 2O^{2-}(s)$$

Each oxygen atom gains two electrons and forms an oxygen ion. Note that because there are two oxygen atoms in $O_2$, four electrons are required.

# Balancing redox reactions

In the examples above, you might have noticed that the two half-equations for the reaction of zinc with copper(II) sulfate can be added up to get the overall equation:

Oxidation:

$$Zn(s) \rightarrow Zn^{2+}(aq) + 2e^-$$

Reduction:

$$Cu^{2+}(aq) + 2e^- \rightarrow Cu(s)$$

Overall reaction:

$$Zn(s) + Cu^{2+}(aq) \rightarrow Zn^{2+}(aq) + Cu(s)$$

Recall that Hess's law involves adding the equations together and cancelling out terms that appear on both sides (see section R1.2.2 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/hesss-law-id-46252/)). The same concept applies here because it all goes back to the law of conservation of mass/energy. Summation of equations, even half reaction equations, will equal the total in terms of mass and energy.

However, that is not always so simple – let's look at the rusting of iron. Consider the two half-reactions and compare them with the overall equation:

Oxidation:

$$Fe(s) \rightarrow Fe^{3+}(s) + 3e^-$$

Reduction:

$$O_2(g) + 4e^- \rightarrow 2O^{2-}(s)$$

Overall reaction:

$$4Fe(s) + 3O_2(g) \rightarrow 2Fe_2O_3(s)$$

In the half-equations above, note that the number of electrons lost by one iron atom(three electrons) does not match the number of electrons received by the two oxygen atoms(four electrons total). Given the electrons gained by the oxygens need to be lost by iron atoms, you need to match the total number of electrons on both equations. To do that, you multiply each of the equations by a factor that will create the same number of total electrons. In this case:

Oxidation:

$$4Fe(s) \rightarrow 4Fe^{3+}(s) + 12e^- \quad \text{(multiply by 4)}$$

Reduction:

$$3O_2(g) + 12e^- \rightarrow 6O^{2-}(s) \quad \text{(multiply by 3)}$$

Now that the total number of electrons matches, you can cancel them out and add up all the reactants and products:

Overall reaction:

$$4Fe(s) + 3O_2(g) \rightarrow 2Fe_2O_3(s) \quad \text{(which would be the } 4Fe^{3+} + 6O^{2-}\text{)}$$

You have now learned how to balance redox reactions in neutral conditions. Try it again with the reaction between magnesium metal and hydrochloric acid.

## Worked example 9

State the oxidation half-reaction for the reaction between magnesium metal and hydrochloric acid.

Oxidation:

$$Mg(s) \rightarrow Mg^{2+}(aq) + 2e^-$$

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Each magnesium atom will lose two electrons and form a magnesium ion.

## Worked example 10

State the reduction half-reaction for the reaction between magnesium metal and hydrochloric acid.

Reduction:

$$HCl(aq) + e^{-} \rightarrow \frac{1}{2}H_2(g) + Cl^{-}(aq)$$

Each hydrogen atom will gain one electron and form half of a diatomic molecule of hydrogen gas.

Because the chloride ion is a spectator ion, you can leave it out of the equation:

$$H^{+}(aq) + e^{-} \rightarrow \frac{1}{2}H_2(g)$$

You can also write the equation without fractional quotients – both are correct:

$$2HCl(aq) + 2e^{-} \rightarrow H_2(g) + 2Cl^{-}(aq)$$

State the overall equation for the reaction between magnesium metal and hydrochloric acid.

Overall reaction:

$$Mg(s) + 2H^{+}(aq) \rightarrow Mg^{2+}(aq) + H_2(g)$$

The half-equations are:

Oxidation:

Kog

$$Mg(s) \rightarrow Mg^{2+}(aq) + 2e^-$$

Reduction:

$$H^+(aq) + e^- \rightarrow \tfrac{1}{2}H_2(g)$$

Because the number of electrons is not the same on both equations, you need to multiply the reduction equation by two:

Reduction:

$$2H^+(aq) + 2e^- \rightarrow H_2(g)$$

Now you can cancel the electrons and add up the remaining equation.

You can also include the spectator ion back, and write the products as $Mg^{2+}(aq) + 2Cl^-(aq)$ or as $MgCl_2(aq)$:

$$Mg(s) + 2HCl(aq) \rightarrow MgCl_2(aq) + H_2(g)$$

Not all reactions happen in neutral conditions. You will also learn to balance redox equations in acidic conditions. What is the overall equation for the oxidation of $Fe^{2+}$ to $Fe^{3+}$ by $Cr_2O_7^{2-}$ (giving $Cr^{3+}$) in an acid solution?

1. Start by writing the oxidation and reduction half-equations and balance all elements except H and O:

   Oxidation:

   $$Fe^{2+}(aq) \rightarrow Fe^{3+}(aq)$$

   Reduction:

   $$Cr_2O_7^{2-}(aq) \rightarrow 2Cr^{3+}(aq)$$

   (note that there are two Cr that need to be balanced)

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2. Assign oxidation states to add the total number of electrons lost/gained in each half-reaction:

   Oxidation:

$$\overset{+2}{Fe^{2+}}(aq) \rightarrow \overset{+3}{Fe^{3+}}(aq) + e^-$$

(oxidation state increases by one)

   Reduction:

$$\overset{+6}{Cr_2O_7^{2-}}(aq) + 6e^- \rightarrow \overset{+3}{2Cr^{3+}}(aq)$$

(oxidation state decreases by 3 × 2Cr)

3. Balance any oxygen atoms by adding water:

   Reduction:

$$Cr_2O_7^{2-}(aq) + 6e^- \rightarrow 2Cr^{3+}(aq) + 7H_2O(l)$$

4. Balance any hydrogen atoms by adding hydrogen ions:

   Reduction:

$$Cr_2O_7^{2-}(aq) + 6e^- + 14H^+(aq) \rightarrow 2Cr^{3+}(aq) + 7H_2O(l)$$

5. Multiply the equations as needed to match the number of electrons on both half-reactions:

   Oxidation:

$$6Fe^{2+}(aq) \rightarrow 6Fe^{3+}(aq) + 6e^-$$

(multiply by 6)

6. Add both half-reactions:

$$6Fe^{2+}(aq) + Cr_2O_7^{2-}(aq) + 14H^+(aq) \rightarrow 6Fe^{3+}(aq) + 2Cr^{3+}(aq) + 7H_2O(l)$$

7. Check your work by counting all atoms on both sides and confirming the total oxidation state on both sides of the equation:

$$\begin{array}{cccccc} +12 & -2 & +14 & \rightarrow\ +18 & +6 & 0 \end{array}$$

$$6Fe^{2+}(aq) + Cr_2O_7^{2-}(aq) + 14H^+(aq) \rightarrow 6Fe^{3+}(aq) + 2Cr^{3+}(aq) + 7H_2O(l)$$

## Worked example 11

Write the balanced equation for this unbalanced reaction:

$$Fe^{2+}(aq) + MnO_4^-(aq) \rightarrow Fe^{3+}(aq) + Mn^{2+}(aq)$$

1. Assign oxidation states to identify oxidised and reduced species:

$$\overset{+2}{Fe^{2+}}(aq) + \overset{+7}{Mn}O_4^-(aq) \rightarrow \overset{+3}{Fe^{3+}}(aq) + \overset{+2}{Mn^{2+}}(aq)$$

Iron(II) ions are oxidised and manganese is reduced.

2. Write the half-equations:

Oxidation:

$$Fe^{2+}(aq) \rightarrow Fe^{3+}(aq) + e^-$$

Reduction:

$$MnO_4^-(aq) + 5e^- \rightarrow Mn^{2+}(aq)$$

3. Balance any oxygen atoms by adding water:

Reduction:

$$MnO_4^-(aq) + 5e^- \rightarrow Mn^{2+}(aq) + 4H_2O(l)$$

4. Balance any hydrogen atoms by adding hydrogen ions:

Reduction:

$$MnO_4^-(aq) + 5e^- + 8H^+(aq) \rightarrow Mn^{2+}(aq) + 4H_2O(l)$$

5. Because the number of electrons is not the same on both equations, you need to multiply the oxidation equation by five:

Oxidation:

$$5Fe^{2+}(aq) \rightarrow 5Fe^{3+}(aq) + 5e^-$$

6. Add both half-reactions:

$$5Fe^{2+}(aq) + MnO_4^-(aq) + 8H^+(aq) \rightarrow 5Fe^{3+}(aq) + Mn^{2+}(aq) + 4H_2O(l)$$

7. Check your work by counting all atoms on both sides and confirming the total oxidation state on both sides of the equation:

$$\underset{5Fe^{2+}(aq)}{+10} + \underset{MnO_4^-(aq)}{-1} + \underset{8H^+(aq)}{+8} \rightarrow \underset{5Fe^{3+}(aq)}{+15} + \underset{Mn^{2+}(aq)}{+2} + \underset{4H_2O(l)}{0}$$

Check your understanding with this question.

## Exercise 1

Click a question to answer

**1.** Determine the correct equation for this reaction in acidic solution.

$$Ag(s) + NO_3^-(aq) \rightarrow Ag^+(aq) + NO(aq)$$

Check answers

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## Making connections

Why are some redox titrations described as 'self-indicating'?

Redox titrations, like acid-base reactions, can be studied through titration. Titration (see section R3.1.8 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/ph-curves-for-strong-acid-base-neutralisation-reactions-id-46057/)) is a laboratory technique that can be used for various purposes such as determining the concentration of a substance in a solution or the amount of a particular substance in a sample, or for measuring the purity of a compound. During a titration, a solution called an indicator is generally used to identify the endpoint of the reaction by a change of colour (see section R3.1.4—5a (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/calculations-involving-ph-and-[h+]-id-44702/)).

Some redox titrations are described as 'self-indicating' because the analyte itself acts as the indicator, without the need for an external indicator solution. In other words, the analyte undergoes a distinct and easily observable change in colour or some other physical property at the endpoint of the titration, indicating the completion of the reaction. One example is the bromine water test for alkenes (see section R3.4.4—5 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/electrophilic-addition-reaction-id-46329/)).

## Activity

- **IB learner profile attribute:**
  - Open-minded
  - Thinker
- **Approaches to learning:**
  - Thinking skills — Applying key ideas and facts in new contexts
  - Communication skills — Using terminology, symbols, and communication conventions consistently and correctly
- **Time required to complete activity:** 30 minutes
- **Activity type:** Individual/pair activity

Now test your knowledge of oxidation and reduction by sorting the terms in **Interactive 3.**

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| Oxidation | Reduction |
| --- | --- |

Gain of electrons

Oxidation state decreases

$Zn^{2+}(aq) + 2e^- \rightarrow Zn(s)$

Oxygen added

Hydrogen added

Oxidation state increases

Loss of electrons

$Zn(s) \rightarrow Zn^{2+}(aq) + 2e^-$

Oxygen removed

Hydrogen removed

Check

**Interactive 3.** Sorting Game: Oxidation and Reduction.